

Meru मेरु

How it's built, and how to build it with us.

In some Eastern traditions, Meru is the cosmic mountain that the sun, moon and stars turn around.

v0.4.18 · for developers · October 2026

github.com/aarora79/meru
Scan for the code

THE PROJECT

A local AI assistant, written in Go, built in the open

Two programs

`merud`, a daemon that keeps models warm and does the work, and `meru`, a thin client over a Unix socket.

Three clients

`meru "…"` for one question, `meru chat` in a terminal, and Meru.app, a Wails v3 window.

Local models

Ollama on loopback runs every model. Meru's code has no path to a cloud model.

Why Go

Single native binaries with no cgo in `merud`, cheap goroutines for a daemon, and code a Python developer can read.

The size

32 internal packages, 6 commands, a coding note per package and 26 notes on Go itself.

The licence

Apache 2.0. Nothing for sale, anywhere.

HOW WE DECIDE

Simple wins every time

- When two designs both work, pick the one with fewer moving parts.
- A function beats a type. A struct beats an interface. An interface appears with its second implementation.
- One way to do each thing: one `dispatch` path, one store, one config file.
- No speculative flags, options or plugin points.
- The standard library first, boring libraries second.
- Add complexity only when a measurement or a second real use asks for it, and say which in the PR.
- Write for a reader new to Go: every package, file and function explains itself.
- `ARCHITECTURE.md` is the contract; change it before the code.

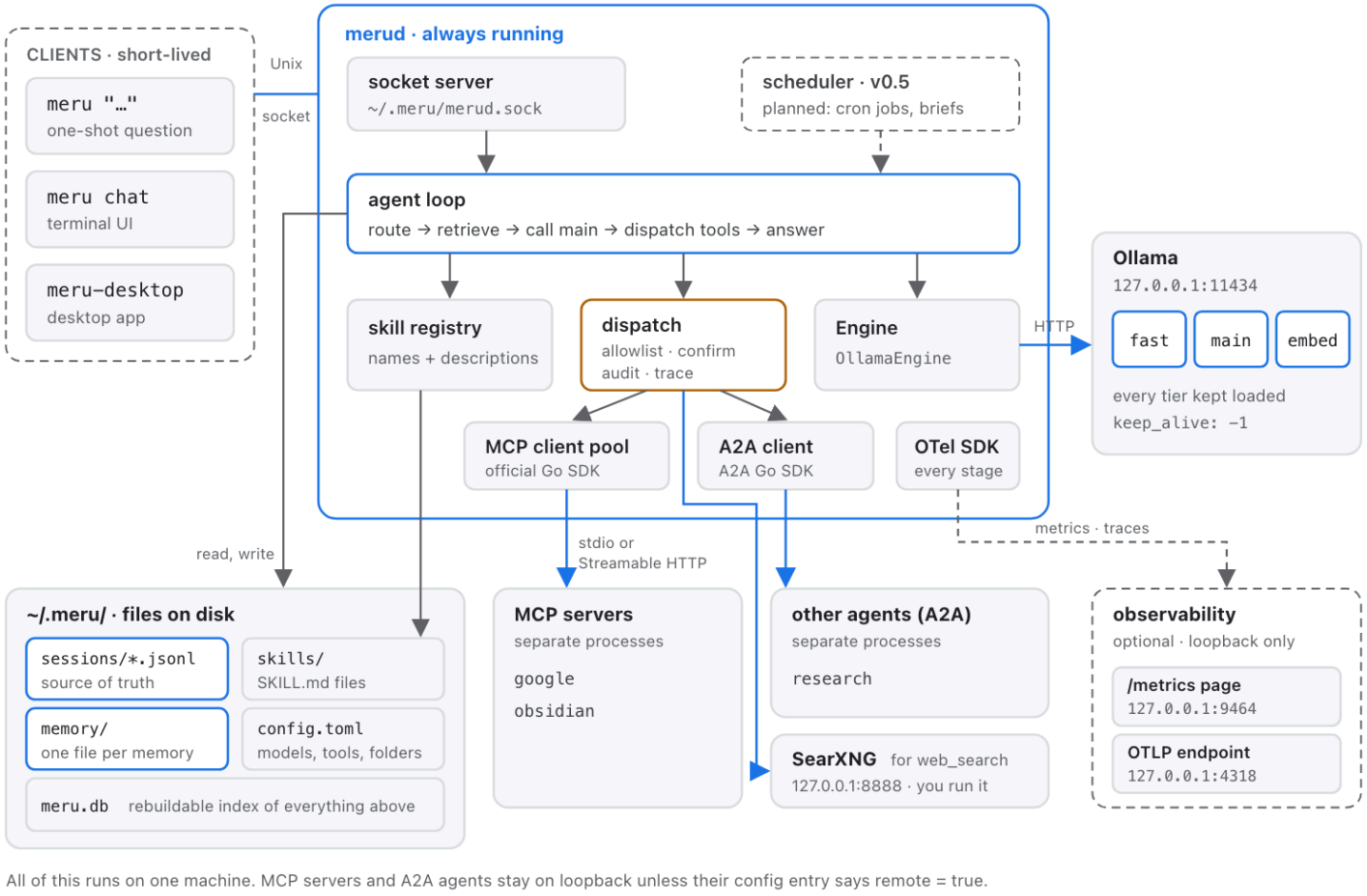
THE RULES THAT DON'T BEND

Five non-negotiables, enforced in CI

Rule	What it means in code
No cloud-model code path	No HTTP client for a model provider, even behind a flag. Model traffic goes to Ollama on loopback.
No telemetry off the machine	OpenTelemetry exports to loopback only; no update checks, no crash reports.
Deny by default	A new MCP server or A2A agent offers nothing until config allowlists it.
Every tool call through `dispatch`	One path logs each call to `tool_calls` and the transcript. Never a second one.
Nothing for sale	No pricing, paid licence or sponsor link anywhere in the project.

`internal/policy` holds the tests: deny-lists of hosts and imports, allowed URLs, and the thin-client rules.

ARCHITECTURE



All of this runs on one machine. MCP servers and A2A agents stay on loopback unless their config entry says remote = true.

A daemon, thin clients and files you own

- `merud` owns the models, the store, the tools and the agent loop.
- Clients hold no model or store logic, and policy tests keep it so.
- Files under `~/.meru` are the truth; `meru.db` is a rebuildable index.

THE HARNESS

One question, end to end, in `merud`

1
Request
One JSON line on the Unix socket: the question, its scope, any pictures.

2
History
The session's last turns, trimmed to fit.

3
Route
The fast model picks one of four routes with one token.

4
Search first
Files, memories and past chats; the web when the question asks for it.

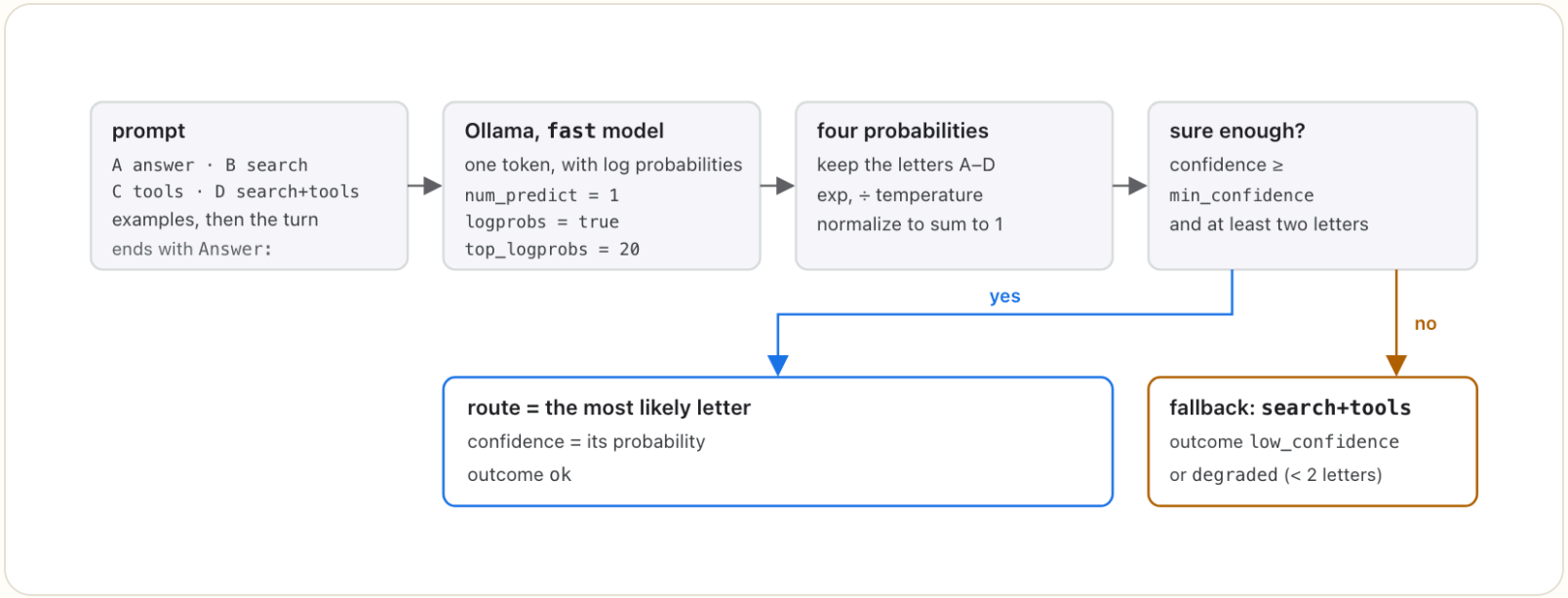
5
Prompt
Built within a budget, stable parts first so Ollama reuses its cache.

6
Rounds
The main model calls tools, up to 8 rounds, the last with none.

7
`dispatch`
Each call allowed, fenced, approved, logged.

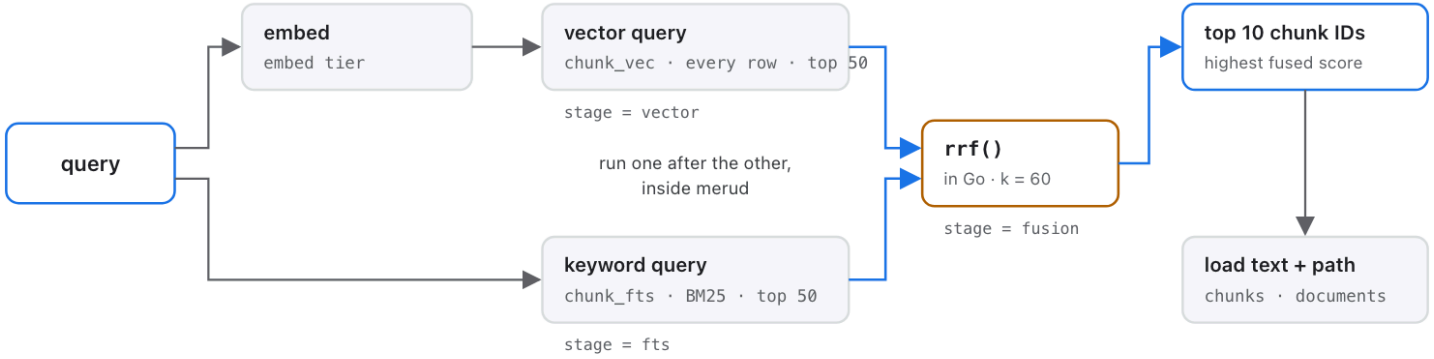
8
Answer
Streams to the client; checks for claims no tool backs; written to the transcript.

A hand-written loop with no agent framework. `docs/architecture/harness.md` walks it function by function.



A router that writes one letter

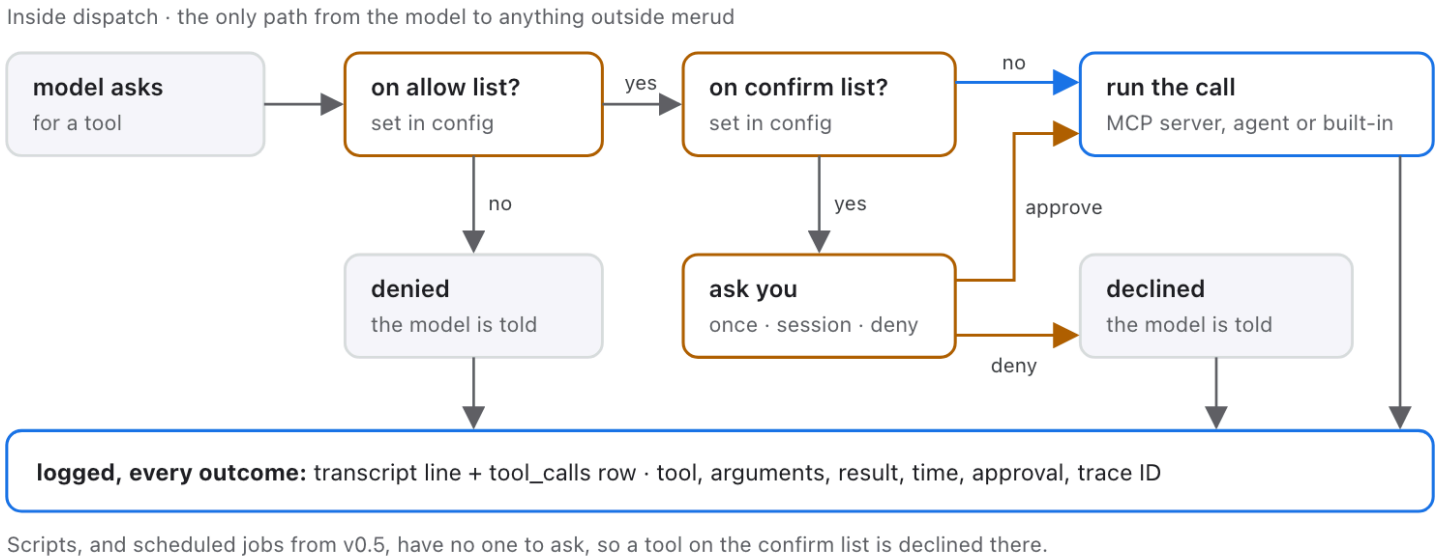
- The fast model labels the four routes A to D and writes one token.
- Ollama's log probabilities give a confidence, so a small model can't garble the choice.
- Labelled questions in `testdata/`; `make router-eval` scores a change.



List sizes (50, 50, 10) are constants in internal/retrieve, not config. Each stage reports its time in meru.retrieval.duration.

Search by meaning and by words

- Vectors and BM25 keyword search, 50 each, merged by reciprocal-rank fusion.
- SQLite through `ncruces/go-sqlite3`: no cgo, with vec1 and FTS5 bundled.
- Text and images in one 768-number space with EmbeddingGemma 2.



`dispatch`: the one path to a tool

- MCP servers, A2A agents, `[[commands]]` and 16 built-in tools.
- Allowlist, scope fence, approval (once, session, deny), audit, metrics.
- An incognito chat still gets a `tool_calls` row, with no arguments or result.

CONNECTORS

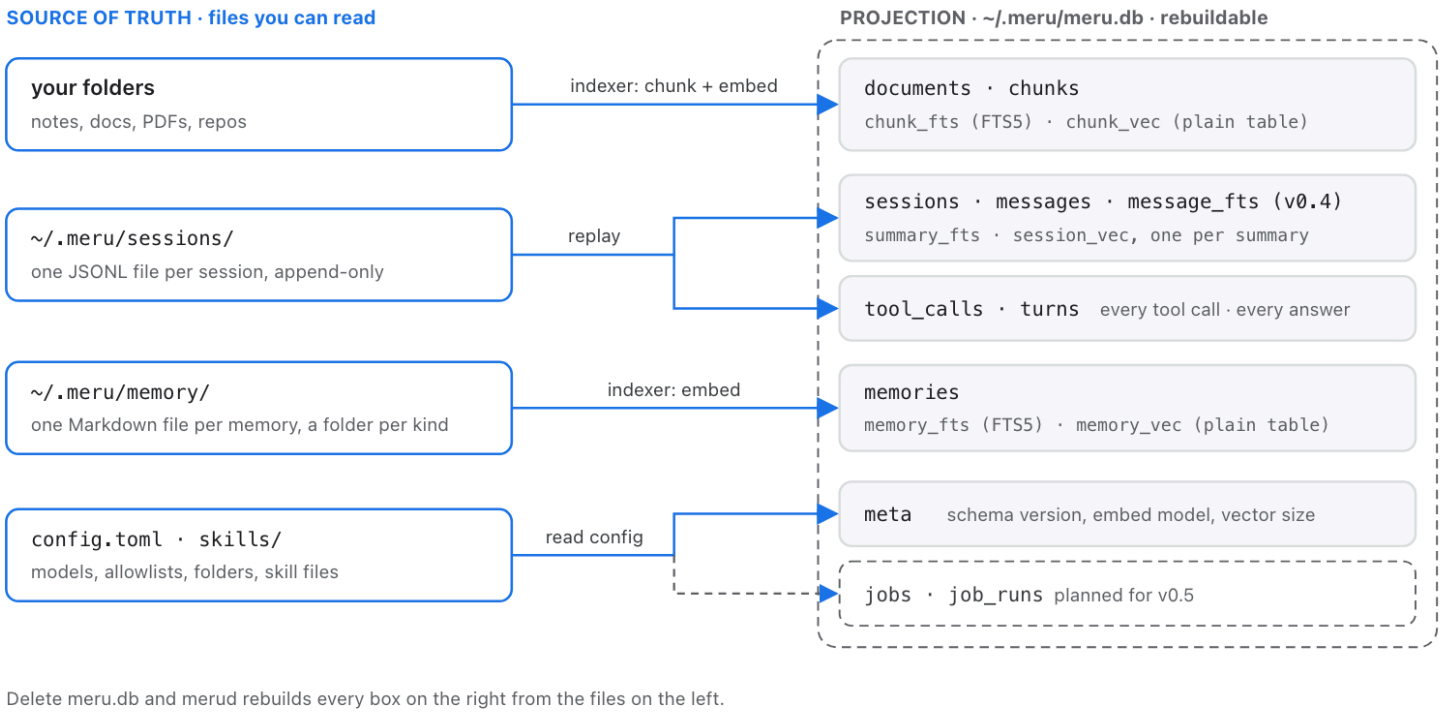
Meru installs and runs the servers it needs

- A manifest per connector, compiled into `merud`: Obsidian, Google, SearXNG, Ollama and Whisper.
- Every pin is exact: package versions, image digests, SHA-256 per download. A policy test refuses `latest`.
- A supervisor per connector installs on enable, starts on use, stops when idle, and backs off after a crash.

Five kinds

- `stdio` and `http`: MCP servers
- `container`: SearXNG in Docker
- `service`: a program speaking plain HTTP, such as `whisper-server`
- `dependency`: Ollama, watched, never installed

STORAGE



Delete meru.db and merud rebuilds every box on the right from the files on the left.

Files are the truth; the database is an index

- JSONL transcripts, Markdown memories, one `config.toml`.
- Delete `meru.db` and Meru rebuilds it.
- A new embedding model drops vectors, keeps text, and re-embeds.

OBSERVABILITY

Measured, and the numbers stay home

- OpenTelemetry metrics and traces for every stage, exported to loopback only; `merud` refuses any other endpoint.
- A `/metrics` page for a Prometheus you run, and a Usage screen in both clients, drawn from `meru.db`.
- Question and answer text stay out of spans unless `capture_content = true`.

Some of the metrics

- `meru.turn.duration` by route, source and outcome
- `meru.route.decisions` : how often the router is unsure
- `meru.tool.calls` by kind, server, tool and outcome
- `meru.retrieval.duration` by stage
- `meru.speech.duration` by model and outcome

Attributes come from small fixed sets: never IDs, paths or text.

A WORKED EXAMPLE

Voice in a day: measure first, then build

The test

- Gemma 4 in Ollama heard well, but at 7.4 GB pushed a model out: about 5 s to a spoken question's first word.
- As a chat model it answered "explain why the sky is blue" instead of writing it down.
- Whisper: 445 to 890 MB in its own process, under half a second for 26 s of speech.

The build, six PRs

- Design in ARCHITECTURE.md first
- CI builds `whisper-server` from pinned source, with a build attestation
- A `service` connector kind, behind a random path per start
- A `transcribe` op, then the microphone in Meru.app

HOW WE WORK

Every change carries its docs and its checks

Contract first

Change `ARCHITECTURE.md`, then carry it into the 300, 200 and 100 pages and their figures.

Explain the code

A PR that changes a package updates its note in `docs/coding-notes/`, or it doesn't merge.

`make check`

Format, vet, staticcheck, race tests, end-to-end tests, five-platform builds, govulncheck, gosec, gitleaks.

Reviewed by personas

The `pr-review` skill reads each PR as security, SRE, Go and architecture reviewers.

Plain prose

Docs, comments and commit messages follow the `writing` skill: short words, active voice.

Releases

`make release` builds, checks and publishes; release notes come from a skill.

CONTRIBUTE

We build in the open, and we're happy to accept contributions

Get going

`git clone https://github.com/aarora79/meru`
`make check`
`go run ./cmd/merud` and `go run ./cmd/meru "hello"`

Read `AGENTS.md`, then `docs/lld.md`, then the coding note for the package you touch. `CONTRIBUTING.md` has the contributor agreement.

From the issue list

- #150 Live transcription
- #84 Meru end to end on Ubuntu
- #85 Meru end to end on Windows
- #77 Python examples of `merud` as an agent harness
- #134 Metrics for the fast and embed model calls

The first outside contribution, #139, is already in review.



Come build with us

Your assistant, your machine, your data.

We build in the open, and we're happy to accept contributions: **see the issue list** on GitHub.

github.com/aarora79/meru
Scan for the code

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